

Jingyu Fu

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EDUCATION

Southern University of Science and Technology (SUSTech)

Sep. 2023 – Jun. 2027 (Expected)

Department of Biomedical Engineering — Major in Intelligent Medical Engineering

Cumulative GPA: 3.84/4.00 **Annual GPA:** 3.94/4.00 **Overall Ranking:** 13 / 49 **Major Ranking:** 1st **Overall Evaluation Ranking:** 3 / 259

Selected Coursework: Machine Learning, Medical Image Processing, Medical Big Data, Signals and Systems, Data Structures and Algorithms, Medical Imaging Systems, Intelligent Sensor Technology, Neuroengineering and BCI, Medical Robotics, Probability Theory, Linear Algebra

Research Interests: Medical Image Analysis, Computer Vision

PUBLICATIONS & PATENTS

Journal & Conference Papers

- Q. Wei, **Jingyu Fu**, et al., “Smartphone Dual-Camera Based Blood Pressure Screening Using Pulse Transit Time between Ear and Finger,” in *Proc. 47th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, 2025. DOI: [10.1109/EMBC58623.2025.11252747](https://doi.org/10.1109/EMBC58623.2025.11252747)
- Jingyu Fu**, Yukai Huang. *VIDEO-LEVEL THYROID ULTRASOUND DIAGNOSIS VIA YOLOV12 M-SIMCLR FUSION AND DYNAMIC KEYFRAME SELECTION*. ChinaXiv publication proof (Certificate No. 202605.00048V1; ChinaXiv: 202605.00048; DOI: 10.12074/202605.00048).
- L. Shang[†], Y. Huang[†], X. Feng, **Jingyu Fu**, et al. *A Pathology Foundation Model-Based Dual-Probability Framework for Risk Stratification and Clinical Triage of Cytologically Indeterminate Thyroid Nodules*. **Modern Pathology** (Resubmitted after revision, Ref: MODPATH-D-26-00161), 2026.

[†] These authors contributed equally to this work.

Patents

- Jingyu Fu** (First Student Inventor), Wenjin Wang, et al., *A Non-contact SpO₂ Monitoring Method and System Based on Smartphone DC Calibration*, Chinese National Invention Patent, No. 202511080860.5, **Granted**.
- Jingyu Fu** (First Student Inventor), et al., *A Deep Learning-based Framework for Automated Thyroid Nodule Detection and Classification in Ultrasound Images*, Chinese National Invention Patent, **Under Review**.

RESEARCH EXPERIENCE

Topic A: Non-contact Physiological Sensing (PPG / BP / SpO₂)

Smartphone Dual-Camera Blood Pressure Screening via PTT

Aug. 2024 – Feb. 2025

Contactless Healthcare Lab, SUSTech — Advisor: Prof. Wenjin Wang — Role: Experimental Lead

- Developed a smartphone-based BP screening pipeline using **ear–finger Pulse Transit Time (PTT)** extracted from synchronized dual-camera rPPG signals, and implemented multi-sensor signal alignment and feature engineering.
- Led the **human-subject experiment design** and data acquisition using an ice-water stimulation protocol; achieved MAE of **3.74 / 2.73 mmHg** (SBP/DBP), meeting AAMI/BHS standards, with results published in **IEEE EMBC 2025** (Second Author).

Multispectral Facial Perfusion and Microcirculation Separation

Oct. 2024 – Feb. 2025

Collab. with Jožef Stefan Institute, Slovenia (Role: Independent Researcher)

- Investigated the separation of **facial macro-perfusion and microcirculation signals** from multispectral RGB videos through spatial–temporal projection.
- Analyzed the feasibility and limitations of depth-resolved physiological signal decoupling under RGB spectral constraints, with a focus on signal coupling across tissue layers.

Non-contact SpO₂ Monitoring via AC–DC Calibration

Feb. 2025 – Jun. 2025

Contactless Healthcare Lab, SUSTech (Role: Primary Inventor)

- Proposed a personalized **AC–DC calibration** strategy using contact PPG (rear camera) to calibrate front-camera DC signals, enhancing robustness against motion and illumination variations.
- Validated the mapping between $R_{oR_{DC}}$ and blood oxygen levels via a breath-holding protocol; filed a **Chinese National Invention Patent** (First Student Inventor).

Topic B: Deep Learning in Thyroid Oncology

AI-Driven Ultrasound Thyroid Nodule Detection & Classification

Sep. 2025 – Present

Collab. with ADA Lab, Hunan University (Role: Independent Researcher)

- Developed a diagnostic pipeline integrating **YOLOv12m** and **SimCLR-based self-supervised learning** to extract stable features from 47k+ unlabeled frames; proposed a **Dynamic Fusion Strategy** (confidence-rule and weighted fusion) to enhance malignant recall to **90.6%**.

- Implemented **DeepSort tracking** and temporal aggregation for patient-level malignancy prediction; built a **Flask+React** platform for video-level visualization and clinical review; filed a **National Invention Patent** and a **First-author** manuscript with **ChinaXiv publication proof** (Certificate No. 202605.00048V1; ChinaXiv: 202605.00048).

Foundation Model for Thyroid FNA Cytology Risk Stratification

Sep. 2025 – Feb. 2026

Collab. with Hangzhou Institute of Medicine, CAS & Hunan University (Role: AI Model Developer)

- Contributed to an AI framework for **risk stratification of indeterminate thyroid nodules (Bethesda III–IV)** using pathology foundation-model features extracted by **TITAN**, and participated in the design and implementation of **DBD-Net** for interpretable 2D risk assessment.
- As an AI model development contributor, I was responsible for PyTorch pipeline setup, TITAN feature extraction implementation, DBD-Net debugging, and five-fold cross-validation experiment reproduction.
- Achieved **AUC 0.948** and **97.9% correct stratification**; manuscript resubmitted after revision to *Modern Pathology* (Ref: MODPATH-D-26-00161), 2026.

INTERNSHIP & PROFESSIONAL EXPERIENCE

Research Intern | Pervasive Interaction Lab, Tsinghua University

July 2025 – Nov. 2025

Project: Continuous NIBP Monitoring via Multi-sensor PTT (Mentor: Jiankai Tang)

Beijing, China

- System Design:** Built an NIBP pipeline via 4-point PPG sensors; synchronized heterogeneous signals (Biopac 2000Hz vs. HUB 111Hz) with time-frequency QC (HR error < 5 BPM). [PI_{Lab}]
- Deployment:** Reproduced the *RingTool* framework and implemented **ONNX-based model deployment** for cross-platform inference; managed remote computing on private cloud servers. [RingTool]

Medical Imaging Research Intern | Heilongjiang Tuomeng Technology Co., Ltd.

July 2026 – Present

Project: Point Cloud-Guided Registration for Longitudinal Lung CT with Non-corresponding Anatomy

Heilongjiang, China

- Problem Framing:** Contributed to longitudinal lung CT registration under disease progression, treatment response, and post-operative scenarios where anatomical consistency is violated.
- Technical Pipeline:** Modeled airway, vascular, and lobar landmarks as semantic point clouds; explored landmark matching, non-correspondence detection, and deformation-field construction for precise local change quantification.

COMPETITIONS & RESEARCH GRANTS

Intelligent Diagnosis of Lower Limb Atherosclerosis via RGB Camera

2024 – 2025

Core Member — Clinical Collab: Shenzhen Third People's Hospital

- Developed a non-contact diagnostic device for **Peripheral Artery Disease (PAD)** using rPPG-based plantar perfusion imaging and multi-dimensional physiological analysis.
- Awards & Grants:** Provincial Third Prize (Challenge Cup), Provincial Funding (Climbing Program), and Provincial Nomination (National Undergraduate Innovation Program).

Other Competition Highlights

- American College Student Mathematical Contest in Modeling (MCM/ICM) — **Meritorious Winner (M Prize)**, 2025.
- China Undergraduate Mathematical Contest in Modeling (CUMCM) — **Provincial First Prize**, 2025.

TECHNICAL SKILLS

Frameworks & Computer Vision	PyTorch, TensorFlow, YOLO, SimCLR, OpenCV, ONNX Deployment
Biomedical Analysis	Multimodal Signal Processing (PPG/ECG), Time-series Modeling
Software & Tools	Python, MATLAB, SOLIDWORKS, Java, Git, Flask, React
Languages	English (CET-4: 627, CET-6: 536, IELTS: 7.0)

AWARDS & HONORS

Dean's Honor Scholarship / Ruoshui Scholarship

2024 – 2025

Sole recipient in Zhiren College (968 students).

SUSTech First-Class Merit Scholarship

2024 – 2025

Ranked 3/259 in comprehensive evaluation (Top 1.2%).

SUSTech Outstanding Student Leader Pacesetter

2024 – 2025

Top 10 students university-wide.

LEADERSHIP & SERVICE

President, Zhiren Student Development Committee

2025 – Present

Led student development and academic support initiatives at the college level; organization recognized as Outstanding Student Organization (First Prize).

President, Zhiren Chinese Orchestra

2024 – 2025

Coordinated large-scale performances and organizational planning; organization recognized as Outstanding Student Organization, and received Outstanding Student Leader.

Student Counselor, Zhiren College

2024 – 2025

Provided peer mentoring and student support services.